

1. Sets

A null or empty set is denoted by $\{ \}$ or ϕ .

$P = Q$ if they have the same elements.

$P \supseteq Q$, Q is subset of P .

$P \subseteq Q$, P is subset of Q .

$P \supset Q$, Q is proper subset of P .

$P \subset Q$, P is proper subset of Q .

$P \cap Q$, Intersection of P and Q .

$P \cup Q$, union of P and Q .

P' complement of P i.e. $\mathcal{E} - P$

2. Simultaneous Equations

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3. Logarithms and Indices

Indices

1. $a^0 = 1$

2. $a^{-p} = \frac{1}{a^p}$

3. $a^{\frac{1}{p}} = \sqrt[p]{a}$

4. $a^{\frac{p}{q}} = (\sqrt[q]{a})^p$

5. $a^m \times a^n = a^{m+n}$

6. $\frac{a^m}{a^n} = a^{m-n}$

7. $(a^m)^n = a^{mn}$

$$8. a^n \times b^n = (ab)^n$$

$$9. \frac{a^n}{b^n} = \left(\frac{a}{b}\right)^n$$

Logarithms

$$1. a^x = y \gg x = \log_a y$$

$$2. \log_a 1 = 0$$

$$3. \log_a a = 1$$

$$4. \log_a xy = \log_a x + \log_a y$$

$$5. \log_a \frac{x}{y} = \log_a x - \log_a y$$

$$6. \log_a b = \frac{\log_c b}{\log_c a}$$

$$7. \log_a b = \frac{1}{\log_b a}$$

$$8. \log_a x^y = y \log_a x$$

$$9. \log_{a^b} x = \log_a x^{\frac{1}{b}}$$

$$10. \log_b x = \log_b c \log_c x = \frac{\log_c x}{\log_c b}$$

4. Quadratic Expressions and Equations

1. Sketching Graph

y-intercept

Put $x=0$

x-intercept

Put $y=0$

Turning pointMethod 1

$$\text{x-coordinate: } x = \frac{-b}{2a}$$

$$\text{y-coordinate: } y = \frac{4ac - b^2}{4a}$$

Method 2

Express $y = ax^2 + bx + c$ as $y = a(x - h)^2 + k$ by completing the square. The turning point is (h, k) .

2. Types of roots of $ax^2 + bx + c = 0$

$$b^2 - 4ac \geq 0 : \text{real roots}$$

$$b^2 - 4ac < 0 : \text{no real roots}$$

$$b^2 - 4ac > 0 : \text{distinct real roots}$$

$$b^2 - 4ac = 0 : \text{equal, coincident or repeated real roots}$$

5. Remainder Factor Theorems

Polynomials

1. $ax^2 + bx + c$ is a polynomial of degree 2.
2. $ax^3 + bx^2 + cx + d$ is a polynomial of degree 3.

Identities

$$P(x) \equiv Q(x) \Leftrightarrow P(x) = Q(x) \quad \text{For all values of } x$$

To find unknowns either substitute values of x , or equate coefficients of like powers of x .

Remainder theorem

If a polynomial $f(x)$ is divided by $(x-a)$, the remainder is $f(a)$

Factor Theorem

$(x-a)$ is a factor of $f(x)$ then $f(a) = 0$

Solution of cubic Equation

- I. Obtain one factor $(x-a)$ by trail and error method.
- II. Divide the cubic equation with a , by synthetic division to find the quadratic equation.
- III. Solve the quadratic equation to find remaining two factors of cubic equation.

For example:

- I. The equation $x^3 + 2x^2 - 5x - 6 = 0$ has $(x-2)$ as one factor, found by trail and error method.
- II. Synthetic division will be done as follows:

$$\begin{array}{r|rrrr}
 2 & 1 & 2 & -5 & -6 \\
 & & 2 & 8 & 6 \\
 \hline
 & 1 & 4 & 3 & 0
 \end{array}$$

↑
remainder

- III. The quadratics equation obtained is $x^2 + 4x + 3 = 0$.
- IV. Equation is solved by quadratic formula, $X=-1$ and $X=-3$.
- V. Answer would be $(x-2)(x+1)(x+3)$.

6. Matrices

1. Order of a matrix

Order of matrix is stated as its number of rows x number of columns. For example, the matrix $\begin{pmatrix} 5 & 6 & 2 \end{pmatrix}$ has order 1×3 .

2. Equality

Two matrices are equal if they are of the same order and if their corresponding elements are equal.

3. Addition

To add two matrices, we add their corresponding elements.

For example, $\begin{pmatrix} 6 & -2 \\ 3 & 5 \end{pmatrix} + \begin{pmatrix} -4 & 2 \\ 4 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 7 & 6 \end{pmatrix}$.

4. Subtraction

To subtract two matrices, we subtract their corresponding elements.

For example, $\begin{pmatrix} 6 & 3 & 5 \\ 9 & 14 & -5 \end{pmatrix} - \begin{pmatrix} 2 & 7 & 5 \\ -4 & 20 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -4 & 0 \\ 12 & -6 & -6 \end{pmatrix}$.

5. Scalar multiplication

To multiply a matrix by k , we multiply each element by k .

For example, $k \begin{pmatrix} 2 & 4 \\ 3 & -1 \end{pmatrix} = \begin{pmatrix} 2k & 4k \\ 3k & -k \end{pmatrix}$ or $3 \begin{pmatrix} 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 6 \\ 12 \end{pmatrix}$.

6. Matrix multiplication

To multiply two matrices, column of the first matrix must be equal to the row of the second matrix. The product will have order row of first matrix X column of second matrix.

For example: $\begin{pmatrix} 2 & 4 \\ 1 & 3 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 3 & 2 & 1 & 4 \\ 1 & 5 & 2 & 7 \end{pmatrix} = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \end{pmatrix}$

To get the first row of product do following:

$$a = (2 \times 3) + (4 \times 1) = 10 \text{ (1}^{\text{st}} \text{ row of first, 1}^{\text{st}} \text{ column of second)}$$

$$b = (2 \times 2) + (4 \times 5) = 24 \text{ (1}^{\text{st}} \text{ row of first, 2}^{\text{st}} \text{ column of second)}$$

$$c = (2 \times 1) + (4 \times 2) = 10 \text{ (1}^{\text{st}} \text{ row of first, 3}^{\text{st}} \text{ column of second)}$$

$$d = (2 \times 4) + (4 \times 7) = 36 \text{ (1}^{\text{st}} \text{ row of first, 4}^{\text{st}} \text{ column of second)}$$

$$e = (1 \times 3) + (3 \times 1) = 6 \text{ (2}^{\text{st}} \text{ row of first, 1}^{\text{st}} \text{ column of second)}$$

$$f = (1 \times 2) + (3 \times 5) = 17 \text{ (2}^{\text{st}} \text{ row of first, 2}^{\text{st}} \text{ column of second)}$$

$$g = (1 \times 1) + (3 \times 2) = 7 \text{ (2}^{\text{st}} \text{ row of first, 3}^{\text{st}} \text{ column of second)}$$

$$h = (1 \times 4) + (3 \times 7) = 25 \text{ (2}^{\text{st}} \text{ row of first, 4}^{\text{st}} \text{ column of second)}$$

$$i = (2 \times 3) + (-1 \times 1) = 5 \text{ (3}^{\text{rd}} \text{ row of first, 1}^{\text{st}} \text{ column of second)}$$

$$j = (2 \times 2) + (-1 \times 5) = -1 \text{ (3}^{\text{rd}} \text{ row of first, 2}^{\text{st}} \text{ column of second)}$$

$$k = (2 \times 1) + (-1 \times 2) = 0 \text{ (3}^{\text{rd}} \text{ row of first, 3}^{\text{rd}} \text{ column of second)}$$

$$l = (2 \times 4) + (-1 \times 7) = 1 \text{ (3}^{\text{rd}} \text{ row of first, 4}^{\text{th}} \text{ column of second)}$$

7. 2 x2 Matrices

a. The matrix $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ is called identity matrix. When it is multiplied with any matrix **X** the answer will be **X**.

b. Determinant of matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ will be $= \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$

c. Adjoint of matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ will be $= \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

d. Inverse of non-singular matrix (determinant is $\neq 0$) $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ will be :

$$\frac{\text{adjoint}}{\text{determinant}} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

8. Solving simultaneous linear equations by a matrix method

$$\begin{matrix} ax + by = h \\ cx + dy = k \end{matrix} \gg \gg \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} h \\ k \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \times \begin{pmatrix} h \\ k \end{pmatrix}$$

7. Coordinate Geometry

Formulas

$$\text{Distance } AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{Midpoint of } AB = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

Parallelogram

If ABCD is a parallelogram then diagonals AC and BD have a common midpoint.

Equation of Straight line

To find the equation of a line of best fit, you need the **gradient(m)** of the line, and the **y-intercept(c)** of the line. The gradient can be found by taking any two points on the line and using the following formula:

$$\text{gradient} = m = \frac{y_2 - y_1}{x_2 - x_1}$$

The y-intercept is the y-coordinate of the point at which the line crosses the y-axis (it may need to be extended). This will give the following equation:

$$y = mx + c$$

Where y and x are the variables, m is the gradient and c is the y-intercept.

Equation of parallel lines

Parallel line have equal gradient.

If lines $y = m_1c_1$ and $y = m_2c_2$ are parallel then $m_1 = m_2$

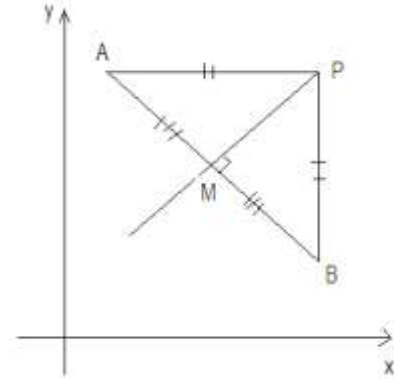
Equations of perpendicular line

If lines $y = m_1c_1$ and $y = m_2c_2$ are perpendicular then $m_1 = -\frac{1}{m_2}$ and $m_2 = -\frac{1}{m_1}$.

Perpendicular bisector

The line that passes through the midpoint of A and B, and perpendicular bisector of AB.

For any point P on the line, $PA = PB$



Points of Intersection

The coordinates of point of intersection of a line and a non-parallel line or a curve can be obtained by solving their equations simultaneously.

8. Linear Law

To apply the linear law for a non-linear equation in variables x and y , express the equation in the form

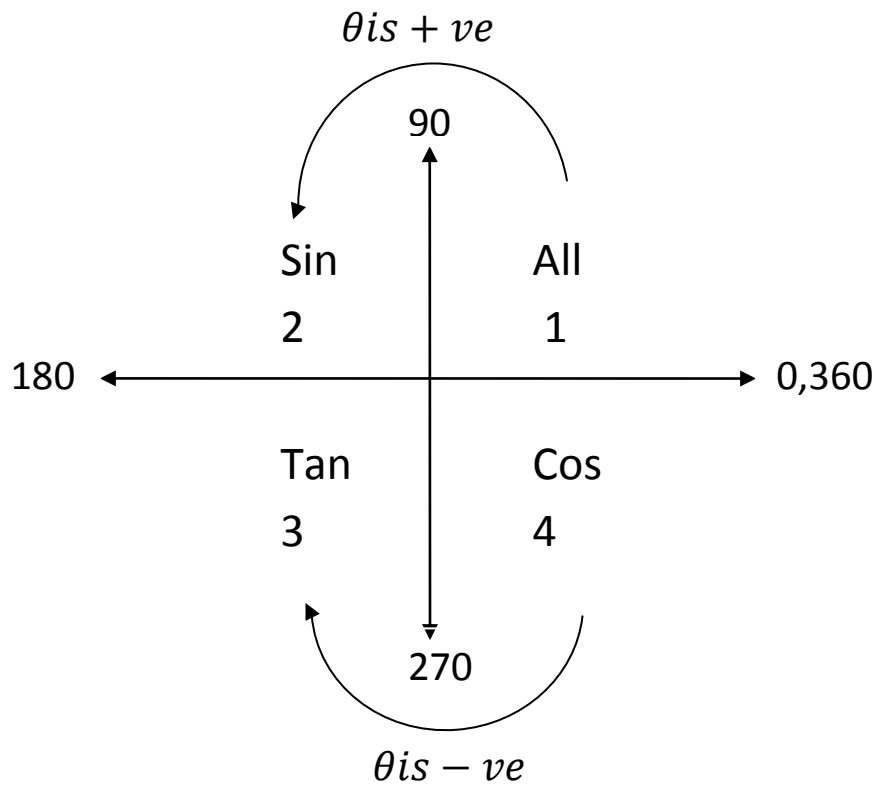
$$Y = mX + c$$

Where X and Y are expressions in x and/or y .

9. Functions

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10. Trigonometric Functions



θ is always acute.

Basics

$$\sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{base}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{perpendicular}}{\text{base}}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Rule 1

$$\sin(90 - \theta) = \cos \theta$$

$$\cos(90 - \theta) = \sin \theta$$

$$\tan(90 - \theta) = \frac{1}{\tan \theta} = \cot \theta$$

Rule 2

$$\sin(180 - \theta) = +\sin \theta$$

$$\cos(180 - \theta) = -\cos \theta$$

$$\tan(180 - \theta) = -\tan \theta$$

Rule 3

$$\sin(180 + \theta) = -\sin \theta$$

$$\cos(180 + \theta) = -\cos \theta$$

$$\tan(180 + \theta) = +\tan \theta$$

Rule 4

$$\sin(360 - \theta) = -\sin \theta$$

$$\cos(360 - \theta) = +\cos \theta$$

$$\tan(360 - \theta) = -\tan \theta$$

Rule 5

$$\sin(-\theta) = -\sin \theta$$

$$\cos(-\theta) = +\cos \theta$$

$$\tan(-\theta) = -\tan \theta$$

Trigonometric Ratios of Some Special Angles

$\cos 45 = \frac{1}{\sqrt{2}}$	$\cos 60 = \frac{1}{2}$	$\cos 30 = \frac{\sqrt{3}}{2}$
$\sin 45 = \frac{1}{\sqrt{2}}$	$\sin 60 = \frac{\sqrt{3}}{2}$	$\sin 30 = \frac{1}{2}$
$\tan 45 = 1$	$\tan 60 = \sqrt{3}$	$\tan 30 = \frac{1}{\sqrt{3}}$

11. Simple Trigonometric Identities

Trigonometric Identities

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

12. Circular Measure

Relation between Radian and Degree

$$\frac{\pi}{2} \text{ radians} = 90^\circ$$

$$\pi \text{ radians} = 180^\circ$$

$$\frac{3\pi}{2} \text{ radians} = 270^\circ$$

$$2\pi \text{ radians} = 360^\circ$$

$s = r\theta$ where s is arc length, r is radius and θ is angle of sector in radians

$$A = \frac{1}{2}rs = \frac{1}{2}r^2\theta \quad \text{where } A \text{ is Area of sector}$$

$$\frac{\text{area of sector}}{\text{area of circle}} = \frac{\text{angle of sector}}{\text{angle of circle}}$$

13. Permutation and Combination

$$n! = n(n-1)(n-2) \times \dots \times 3 \times 2 \times 1$$

$$0! = 1$$

$$n! = n(n-1)!$$

$${}_nP_r = \frac{n!}{(n-r)!}$$

$${}_nC_r = \frac{n!}{(n-r)!r!}$$

14. Binomial Theorem

$$(a+b)^n = a^n + {}^nC_1 a^{n-1}b + {}^nC_2 a^{n-2}b^2 + {}^nC_3 a^{n-3}b^3 + \dots + b^n$$

$$T_{r+1} = {}^nC_r a^{n-r} b^r$$

15. Differentiation

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(ax^m + bx^n) = amx^{m-1} + bnx^{n-1}$$

$$\frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Where 'v' and 'u' are two functions

Gradient of a curve at any point P(x,y) is $\frac{dy}{dx}$ at x

16. Rate of Change

The rate of change of a variable x with respect to time is $\frac{dx}{dt}$

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

$$\frac{\delta y}{\delta x} \approx \frac{dy}{dx}$$

$$\text{Percentage change in } x = \frac{\delta x}{x} \times 100\%$$

$$f(x + \delta x) = y + \delta y \approx y + \frac{dy}{dx} \delta x$$

17. Higher Derivative

$\frac{dy}{dx} = 0$ when $x = a$ then point $(a, f(a))$ is a stationary point.

$\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} \neq 0$ when $x = a$ then point $(a, f(a))$ is a turning point.

For a turning point T

- I. If $\frac{d^2y}{dx^2} > 0$, then T is a minimum point.

II. If $\frac{d^2y}{dx^2} < 0$, then T is a maximum point.

18. Derivative of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sin^n x) = n \sin^{n-1} x \cos x$$

$$\frac{d}{dx}(\cos^n x) = -n \cos^{n-1} x \sin x$$

$$\frac{d}{dx}(\tan^n x) = n \tan^{n-1} x \sec^2 x$$

19. Exponential and Logarithmic Functions

$$\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$$

$$\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$$

A curve defined by $y=\ln(ax+b)$ has a domain $ax+b>0$ and the curve cuts the x-axis at the point where $ax+b=1$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\ln u) = \frac{1}{u} \frac{du}{dx}$$

$$\frac{d}{dx}[\ln(ax + b)] = \frac{a}{ax + b}$$

20.Integration

$$\frac{dy}{dx} = x \Leftrightarrow y = \int x \, dx$$

$$\frac{d}{dx}\left(\frac{1}{2}x^2 + c\right) = x \Leftrightarrow \int x \, dx = \frac{1}{2}x^2 + c$$

$$\int ax^n \, dx = \frac{ax^{n+1}}{n+1} + c$$

$$\int (ax^n + ab^m) \, dx = \frac{ax^{n+1}}{n+1} + \frac{bx^{m+1}}{m+1} + c$$

$$\int (ax + b)^n \, dx = \frac{(ax + b)^{n+1}}{a(n+1)} + c$$

$$\frac{d}{dx}[F(x)] = f(x) \Leftrightarrow \int_a^b f(x) \, dx = F(b) - F(a)$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_a^a f(x) dx = 0$$

$$\frac{d}{dx}(\sin x) = \cos x \Leftrightarrow \int \cos x dx = \sin x + c$$

$$\frac{d}{dx}(-\cos x) = \sin x \Leftrightarrow \int \sin x dx = -\cos x + c$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \Leftrightarrow \int \sec^2 x dx = \tan x + c$$

$$\frac{d}{dx}\left[\frac{1}{a}\sin(ax+b)\right] = \cos(ax+b) \Leftrightarrow \int \cos(ax+b) dx = \frac{1}{a}\sin(ax+b) + c$$

$$\frac{d}{dx}\left[-\frac{1}{a}\cos(ax+b)\right] = \sin(ax+b) \Leftrightarrow \int \sin(ax+b) dx = -\frac{1}{a}\cos(ax+b) + c$$

$$\frac{d}{dx}\left[\frac{1}{a}\tan(ax+b)\right] = \sec^2(ax+b) \Leftrightarrow \int \sec^2(ax+b) dx = \frac{1}{a}\tan(ax+b) + c$$

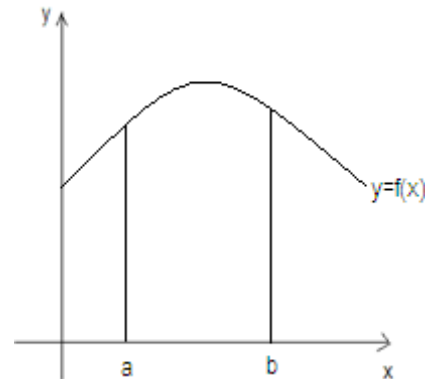
$$\frac{d}{dx}(e^x) = e^x \Leftrightarrow \int e^x dx = e^x + c$$

$$\frac{d}{dx}(-e^{-x}) = e^{-x} \Leftrightarrow \int e^{-x} dx = -e^{-x} + c$$

21.Applications of Integration

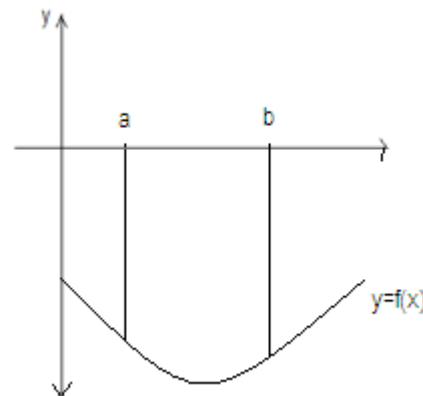
For a region R above the x-axis, enclosed by the curve $y=f(x)$, the x-axis and the lines $x=a$ and $x=b$, the area R is:

$$A = \int_a^b f(x) dx$$



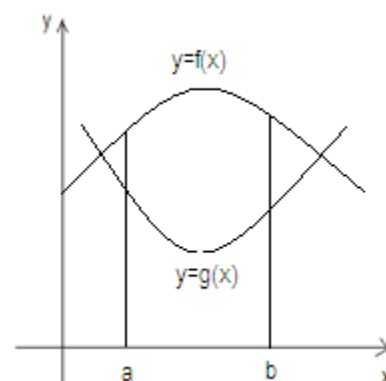
For a region R below the x-axis, enclosed by the curve $y=f(x)$, the x-axis and the lines $x=a$ and $x=b$, the area R is:

$$A = \int_a^b -f(x) dx$$



For a region R enclosed by the curves $y=f(x)$ and $y=g(x)$ and the lines $x=a$ and $x=b$, the area R is:

$$A = \int_a^b [f(x) - g(x)] dx$$



22. Kinematics

$$v = \frac{ds}{dt}$$

$$a = \frac{dv}{dt}$$

$$s = \int v \, dt$$

$$v = \int a \, dt$$

$$\text{Average speed} = \frac{\text{total distance travelled}}{\text{total time taken}}$$

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$s = \frac{1}{2}(u + v)t$$

$$v^2 = u^2 + 2as$$

23. Vectors

If $\overrightarrow{OP} = \begin{pmatrix} x \\ y \end{pmatrix}$ then $|\overrightarrow{OP}| = \sqrt{x^2 + y^2}$

$\mathbf{b} = k\mathbf{a}$ and $k > 0$ $\mathbf{a}$ and $\mathbf{b}$ are in the same direction

$\mathbf{b} = k\mathbf{a}$ and $k < 0$ $\mathbf{a}$ and $\mathbf{b}$ are opposite in direction

Vectors expressed in terms of two parallel vectors $\mathbf{a}$ and $\mathbf{b}$:

$$p\mathbf{a} + q\mathbf{b} = r\mathbf{a} + s\mathbf{b} \Leftrightarrow p = r \text{ and } q = s$$

If A, B and C are collinear points $\Leftrightarrow AB=kBC$

If P has coordinates (x, y) in a Cartesian plane, then the position vector of P is

$$\overrightarrow{OP} = x\mathbf{i} + y\mathbf{j}$$

where $\mathbf{i}$ and $\mathbf{j}$ are unit vectors in the positive direction along the x-axis and the y-axis respectively.

Unit vector in the direction of $\overrightarrow{OP}$ is

$$\frac{1}{\sqrt{x^2 + y^2}}(x\mathbf{i} + y\mathbf{j}) \text{ or } \frac{1}{\sqrt{x^2 + y^2}}\begin{pmatrix} x \\ y \end{pmatrix}$$

24. Relative velocity
